ESE 3203 Concept Primer

Petroleum and Natural Gas Processing · KUET · read this first: the subject from zero, before the past-paper solutions

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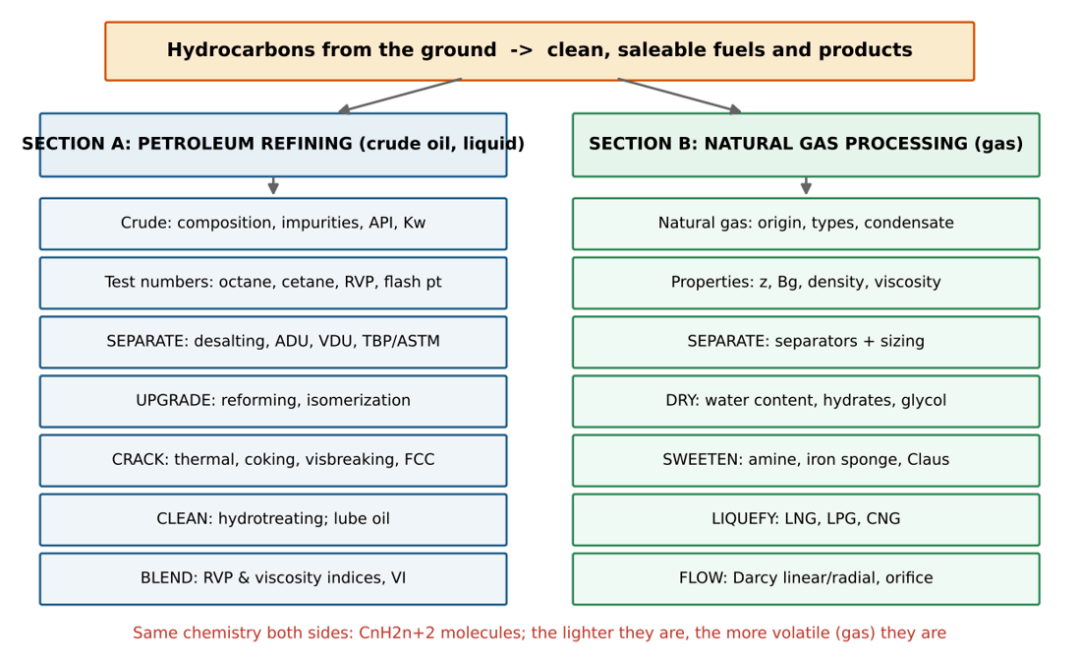
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How to use this primer

This booklet teaches petroleum and natural gas processing from zero, in the order the exam needs it. Read it once before the Complete Solutions. Each section has a green **core idea** box (learn this first), the key figure, the formulas with every symbol and unit, and a short worked example. Most of this course is **descriptive**: you score by drawing the process diagram and explaining each step in order. The numericals are few and repeat every year (API, viscosity blending, VI, gas properties, Campbell water content, separator sizing, Darcy flow), so they are easy marks once you know the steps.

The pass set (from the Question Map): refining A1-A7 and natural gas B1-B6; B7 (Darcy flow and orifice meter) is the next to add. Question numbers move (in 2021 the two sections were swapped), so learn by topic.

The whole course on one page



Concept map of ESE 3203

Read the map as two parallel factories. The **refinery** (Section A) takes liquid crude oil, separates it by boiling point, converts the heavy and low-octane parts into valuable fuels, cleans out sulfur and blends the products. The **gas plant** (Section B) takes the gas stream from the well, separates liquids, removes water and acid gases, and either pipes the gas or liquefies it (LNG, LPG (liquefied petroleum gas)).

Units and maths refresher

> The oil and gas industry uses **field units**: pressure in psi, temperature in °F or °R, volume in barrels and cubic feet, flow in bbl/day or MMscfd. Learn the conversions once and write the unit on every line.

Quantity	Conversion / rule
absolute temperature	°R = °F + 460 (459.67); K = °C + 273.15; °F = 1.8°C + 32
absolute pressure	psia = psig + 14.7; 1 atm = 14.696 psi = 101.325 kPa = 760 mmHg
volume	1 bbl (barrel) = 42 US gal = 5.615 ft^3 = 0.159 m^3; 1 m^3 = 35.31 ft^3 = 6.29 bbl
gas volume	scf = standard cubic foot at 14.7 psia and 60 °F; MMscf = 10^6 scf; MMscfd = 10^6 scf per day
area	1 acre = 43,560 ft^2
density	ρ = SG × 1000 kg/m^3 = SG × 62.4 lbm/ft^3
viscosity	1 cP = 10^{-3} Pa·s; kinematic 1 cSt = 1 mm^2/s; $\nu = \mu/\rho$
permeability	1 darcy = 1000 md (millidarcy) ≈ 0.987 × 10^{-12} m^2

Logarithms (needed for blending and radial flow)

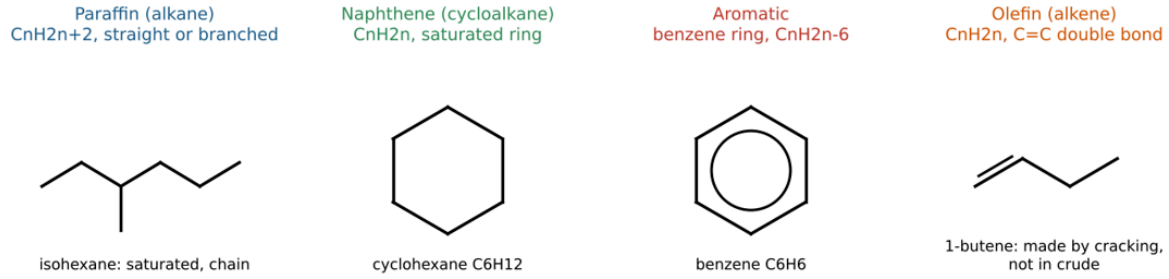
- $\log_{10} 100 = 2$ because $10^2 = 100$; to undo a log, raise 10 to it: if log v = 2.0997 then $v = 10^{2.0997} = 125.8$.
- ln (natural log) uses base e = 2.718: $\ln(r/r_w)$ grows slowly, so pressure changes fast near the well and slowly far away.
- Powers: $12^{1.25} = 22.3$ (calculator: 12 ^ 1.25); to undo $x^{1.25}$, raise to 1/1.25 = 0.8.

Mole-fraction averages (Kay's rule)

For a gas mixture, a property of the mixture is often estimated as $\Sigma\, y_i P_i$, where y_i is the mole fraction (mol% ÷ 100). Make a table with columns y, M, p_c, T_c and the three products; sum each product column. Check that the y column adds to 1.000.

Crude oil and hydrocarbons (A1)

> Crude oil is a mixture of hydrocarbons (only C and H) of many sizes and shapes, plus small amounts of S, N, O, metals, salt and water. **Size** decides boiling point (bigger = higher boiling). **Shape** decides quality: branched chains and benzene rings resist engine knock; straight chains ignite easily (good for diesel, bad for petrol).



The four hydrocarbon families

Family	General formula	Saturated?	Key quality
Paraffin (alkane)	C_nH_{2n+2}	yes (all single bonds)	n-paraffins: high cetane, low octane; iso-paraffins: high octane
Naphthene (cycloalkane)	C_nH_{2n}	yes, ring	medium octane; turned into aromatics by reforming
Aromatic	C_nH_{2n-6} (benzene ring)	no (ring with alternating double bonds)	very high octane; low cetane; smoke
Olefin (alkene)	C_nH_{2n}	no, C=C	made by cracking; high octane but unstable (gum)

API gravity measures how light an oil is:

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API = 141.5/SG - 131.5
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Symbol	Meaning	Unit
`SG`	specific gravity of the oil at 60 °F (density ÷ density of water)	–
`API`	American Petroleum Institute gravity (water = 10)	°API

Example: SG 0.85 → 141.5/0.85 = 166.47 → API = 35.0 (a light/medium crude). Lighter oil = higher API = more gasoline = more valuable.

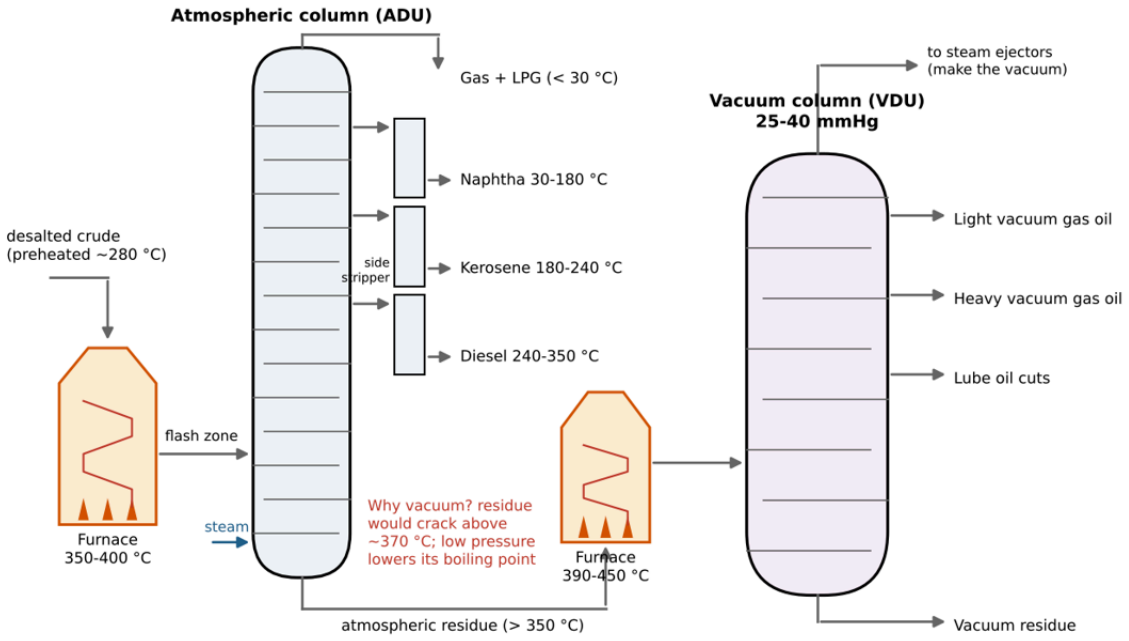
Fuel quality numbers (A2)

A petrol (spark-ignition) engine needs fuel that does **not** self-ignite when compressed: high **octane number**. A diesel (compression-ignition) engine needs fuel that **does** self-ignite quickly: high **cetane number**. Volatility is measured by distillation, RVP (Reid vapour pressure) and flash point.

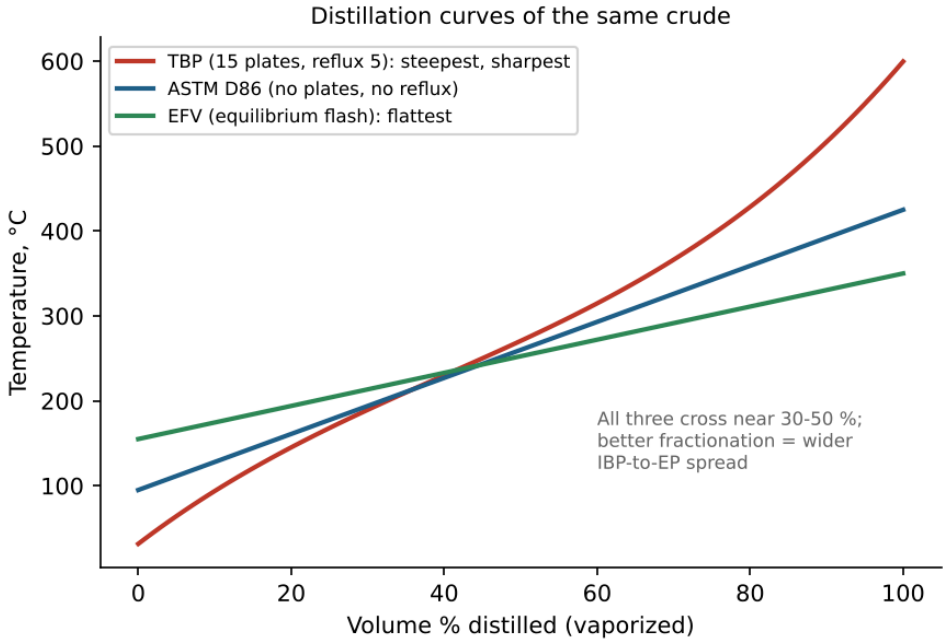
Number	Reference fuels (100 / 0)	High value means
Octane (RON, MON (motor octane number))	iso-octane / n-heptane	resists knock; allows higher compression ratio
Cetane	n-cetane ($C_{16}H_{34}$) / α-methylnaphthalene	short ignition delay; smooth diesel combustion
RVP (psi)	–	evaporates easily: easy cold start, but vapour lock and losses if too high
Flash point (°C)	–	safer to store (needs higher temperature to flash)

Separating crude: distillation and desalting (A3)

Distillation separates by boiling point. The crude is first **desalted** (washed with water, then the water is removed in an electric field), heated in a furnace, and flashed into the **atmospheric column**. The heavy residue that would crack if heated further is distilled in a **vacuum column**, where low pressure lowers boiling points.



Atmospheric and vacuum distillation



Distillation curves: TBP (true boiling point) sharpest, EFV (equilibrium flash vaporization) flattest

Stokes' law explains settling in the desalter:

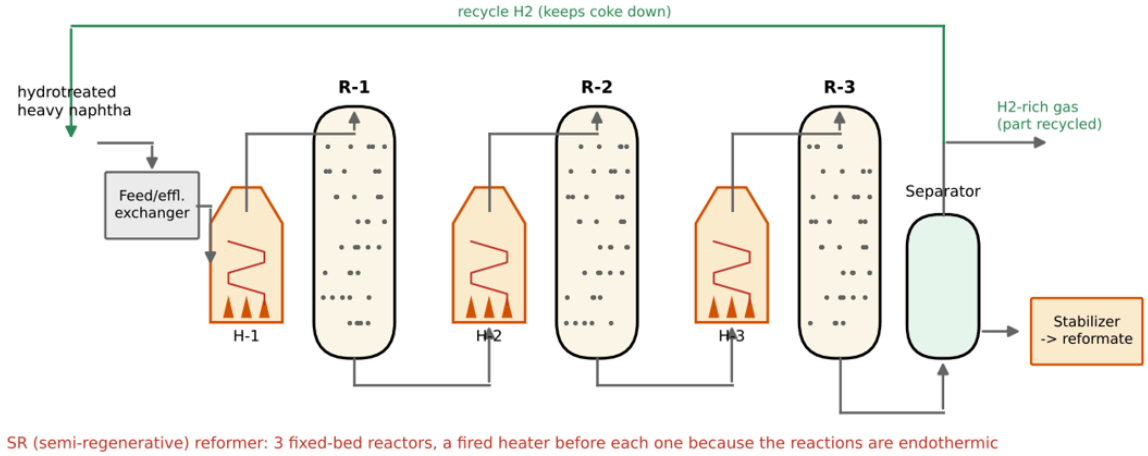
$$v = g\, d^2(\rho_w - \rho_o)/(18\mu_o)$$

Symbol	Meaning	Unit
v	settling velocity of a water drop	m/s
d	drop diameter	m
ρ_w, ρ_o	density of water and oil	kg/m^3
μ_o	oil viscosity	Pa·s

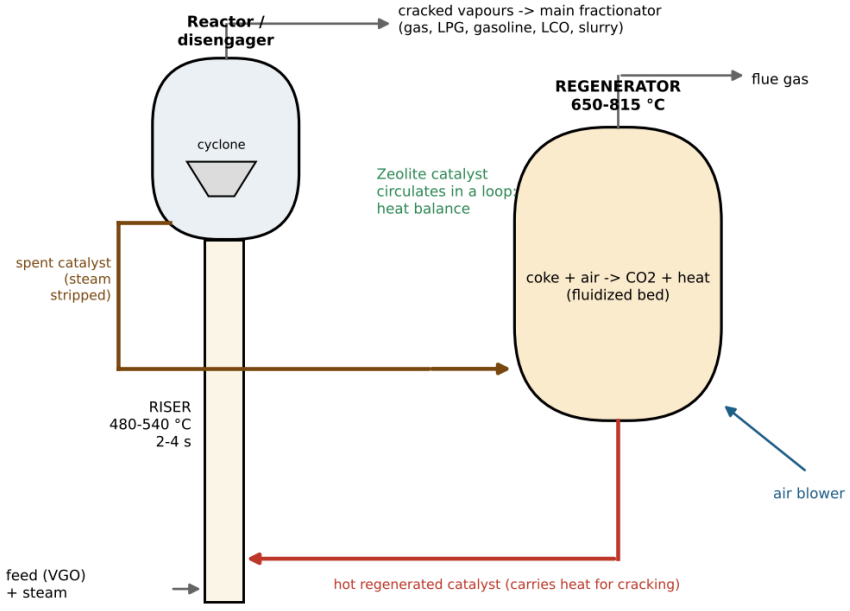
So the desalter makes drops bigger (electric field, d^2) and the oil thinner (heating lowers μ): both raise v.

Changing molecules: reforming, isomerization, cracking (A4, A5)

Reforming rearranges naphtha molecules into aromatics (octane up, H_2 released) over a platinum catalyst; reactions absorb heat, so the gas is reheated between reactors. **Isomerization** turns straight C_5/C_6 chains into branched ones. **Cracking** breaks big molecules into small ones: thermally (coking, visbreaking) or with a catalyst (FCC) or with catalyst + hydrogen (hydrocracking).

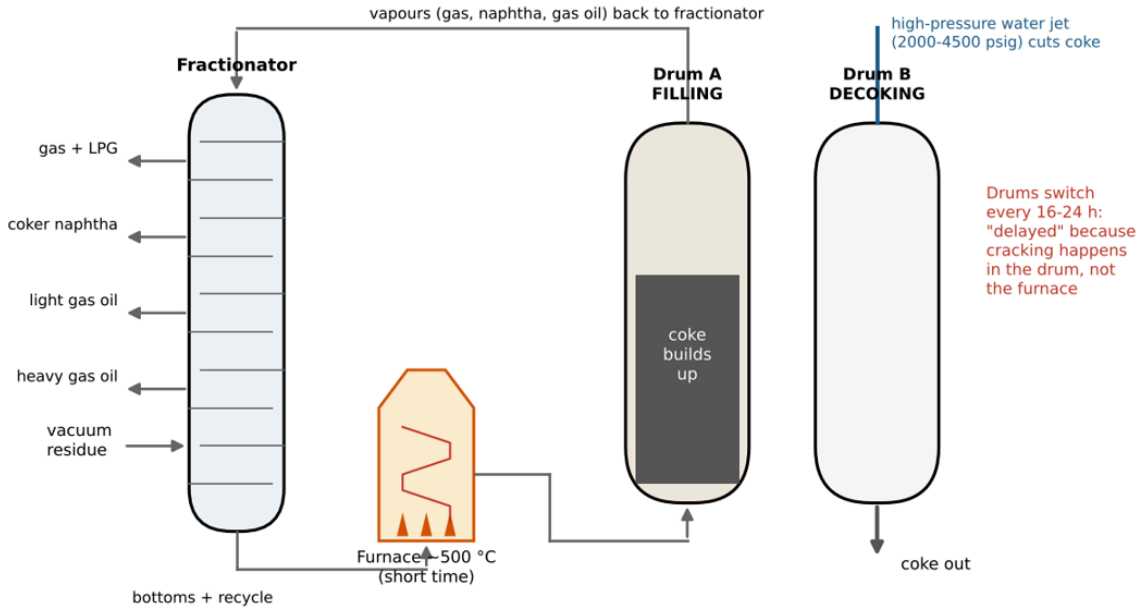


Semi-regenerative reformer



FCC (fluid catalytic cracking): riser and regenerator in a heat-balanced loop

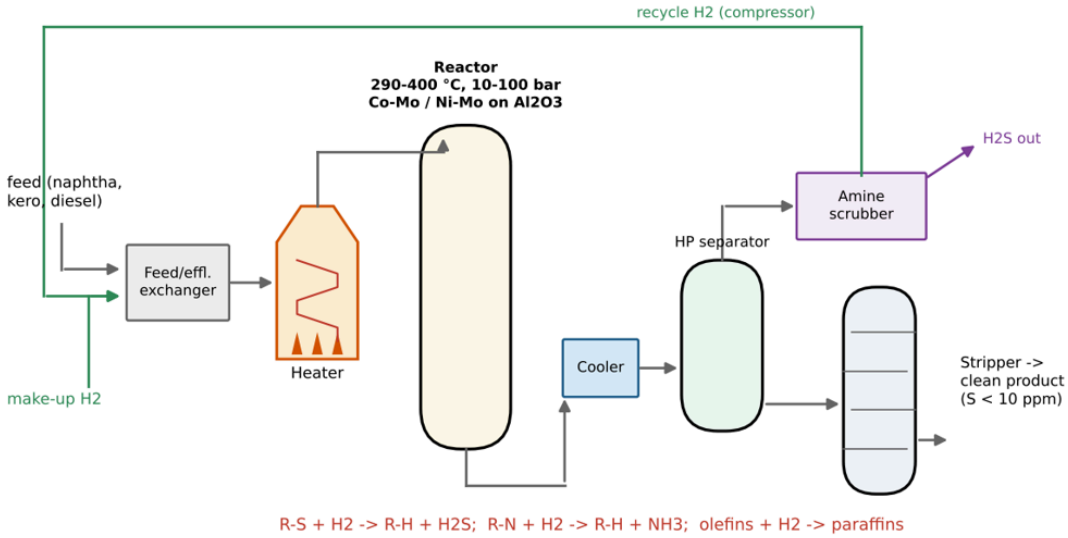
Endothermic = absorbs heat (temperature falls through the bed); **exothermic** = releases heat. A **bifunctional catalyst** has a metal site (Pt: adds/removes hydrogen) and an acid site (alumina, zeolite: rearranges and breaks carbon chains).



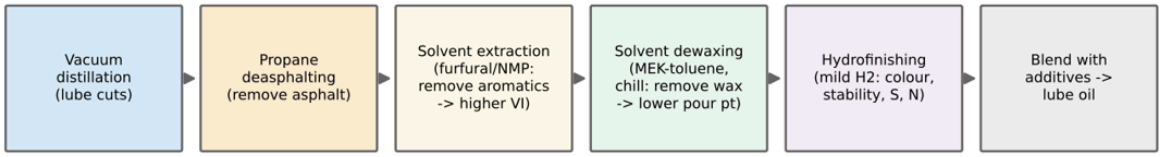
Delayed coker: cracking finishes in the drum

Cleaning with hydrogen and making lube oil (A6)

Hydrotreating: hydrogen over a Co-Mo or Ni-Mo catalyst turns sulfur into H_2S and nitrogen into NH_3, which are washed out; the fuel is left clean and stable. **Lube oil** is made by removing asphalt, aromatics and wax from vacuum distillates, then polishing with hydrogen.



Hydrotreater



Lube oil chain

Blending and viscosity index (A7)

> Some properties mix like an average (SG, sulfur). Others do not (viscosity, RVP, flash point). For those, convert each component to a **blending index**, average the indices by volume fraction, then convert back.

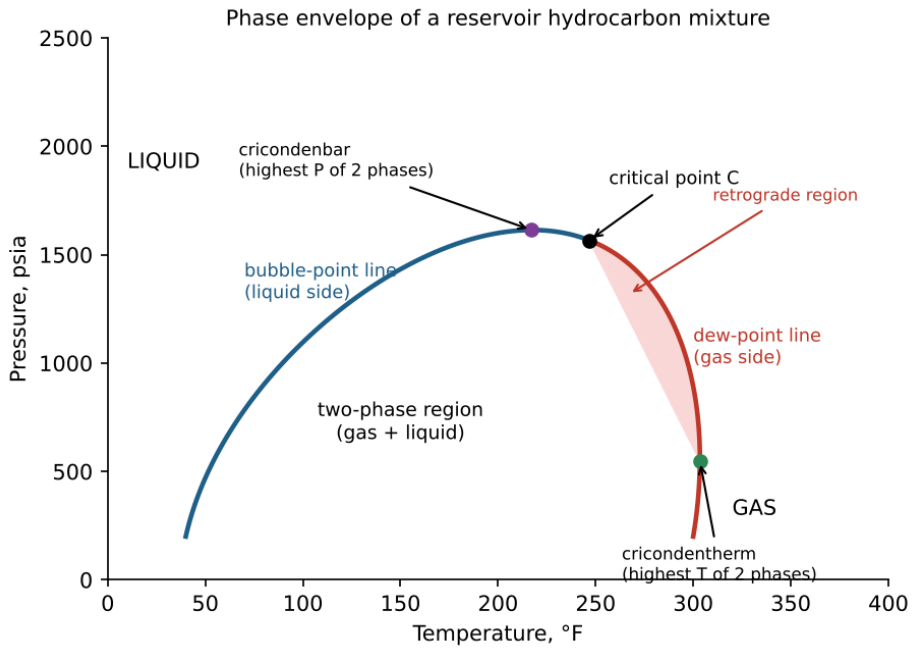
$x_i = V_i/\Sigma V$; $BI_{vis} = logv/(3 + logv)$; $v_{blend} = 10^{3BI/(1-BI)}$; $BI_{RVP} = RVP^{1.25}$; $VI = 100(L - U)/(L - H)$

Symbol	Meaning	Unit
x_i	volume fraction of component i	–
v	kinematic viscosity	cSt
RVP	Reid vapour pressure	psi
U	viscosity of the test oil at 40 °C	cSt
L, H	40 °C viscosity of VI-0 and VI-100 reference oils with the same 100 °C viscosity as the test oil	cSt

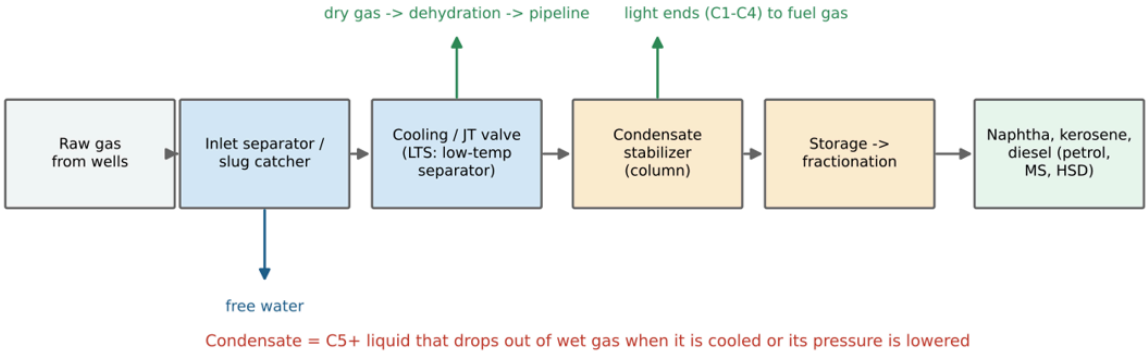
Example: equal volumes of 50 cSt and 200 cSt oil. BI = 1.699/4.699 = 0.3616 and 2.301/5.301 = 0.4341; average 0.3978; logv = 3 × 0.3978/0.6022 = 1.982; v = 95.9 cSt (not the simple average 125).

Natural gas basics and the phase envelope (B1)

> Natural gas is mainly methane. Wet gas carries heavier components that condense to **condensate** when cooled. The **phase envelope** (p-T diagram) shows where a mixture is liquid, gas or both; gas-condensate reservoirs lie between the critical point and the cricondentherm, where lowering pressure makes liquid (retrograde condensation).



Phase envelope



Condensate separation train

Real gas properties (B2)

> Real gas: pV = znRT. z is read from the Standing-Katz chart using **pseudo-reduced** pressure and temperature (p/p_{pc}, T/T_{pc}), where p_{pc}, T_{pc} are mole-fraction averages of the critical constants. Sour gas needs the Wichert-Aziz correction.

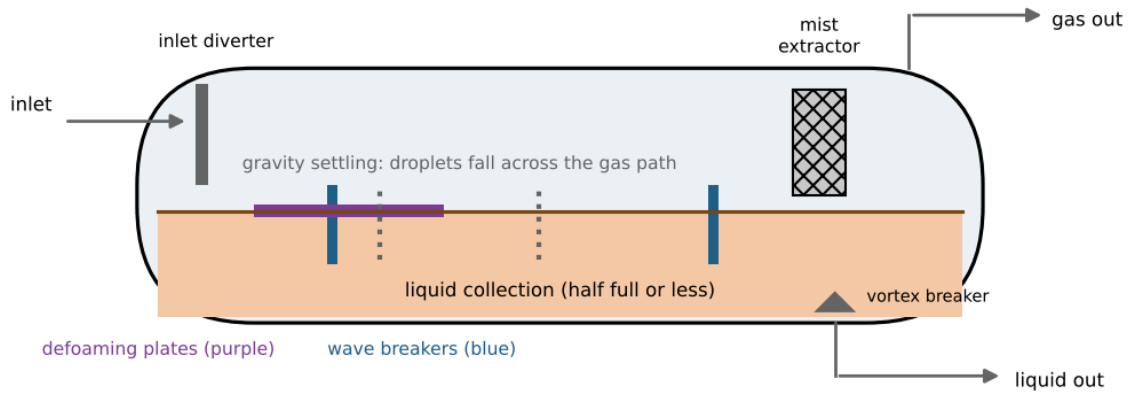
$\rho_g = pM_a/(zRT)$; $B_g = 0.02827\ zT/p$; $\gamma_g = M_a/28.97$; $c_g = 1/p - (1/z)(dz/dp)$

Symbol	Meaning	Unit
p	absolute pressure	psia
T	absolute temperature	°R
z	gas compressibility (deviation) factor	–
M_a	apparent molecular weight $\Sigma y_i M_i$	lb/lb-mol
R	gas constant 10.73	psia·ft^3/(lb-mol·°R)
B_g	gas formation volume factor	ft^3/scf
c_g	isothermal gas compressibility	psi^{-1}
φ	porosity	fraction
k	permeability	md

Example: methane (M = 16.04) at 1000 psia, 100 °F (560 °R), z = 0.89: ρ = 1000 × 16.04/(0.89 × 10.73 × 560) = 3.00 lbm/ft^3; B_g = 0.02827 × 0.89 × 560/1000 = 0.0141 ft^3/scf.

Separators (B3)

> A separator slows the well stream down (inlet diverter), gives droplets time to fall (settling section), catches fine mist (mist extractor) and holds liquid long enough for gas to escape (liquid section). Size it for gas (diameter, Souders-Brown) and for liquid (retention volume), and take the larger.



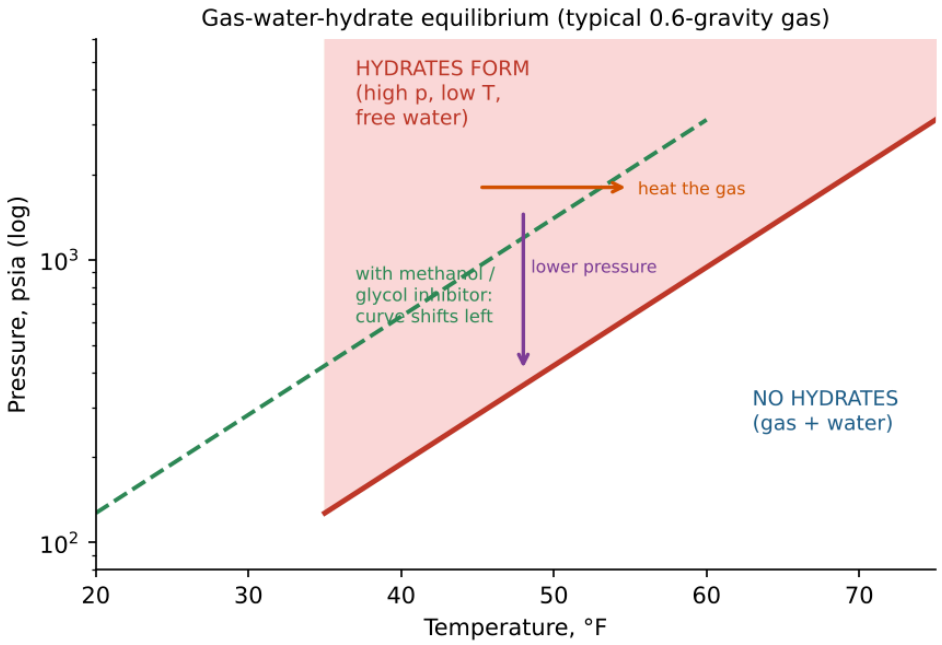
Horizontal separator

$$D^2 = q_g Z(T + 460)\rho_g^{0.5}/[2.40Kp(\rho_L - \rho_g)^{0.5}]; \quad V_L = Wt/1440; \quad V_L = 0.1399D^2h \text{ (vertical)}$$

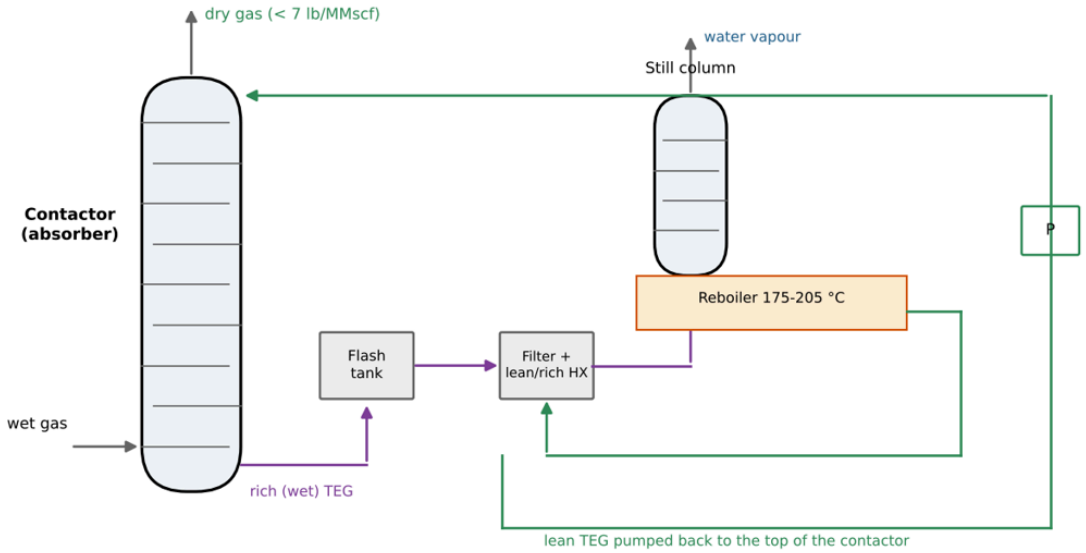
Symbol	Meaning	Unit
D	separator diameter	ft
q_g	gas rate	MMscfd
T	temperature	°F
K	Souders-Brown constant (0.167 vertical, 0.382 horizontal, 0.35 sphere)	–
W	liquid rate = GLR (gas-liquid ratio) × q_g	bbl/day
t	retention time	min
V_L	liquid settling volume	bbl

Water, hydrates and dehydration (B4)

Gas from the well is saturated with water. If it cools under pressure, liquid water and gas form **hydrates** (solid). Remove the water with glycol (absorption) or solid desiccants (adsorption); or keep the gas warm, lower the pressure, or inject methanol/glycol.



Hydrate curve

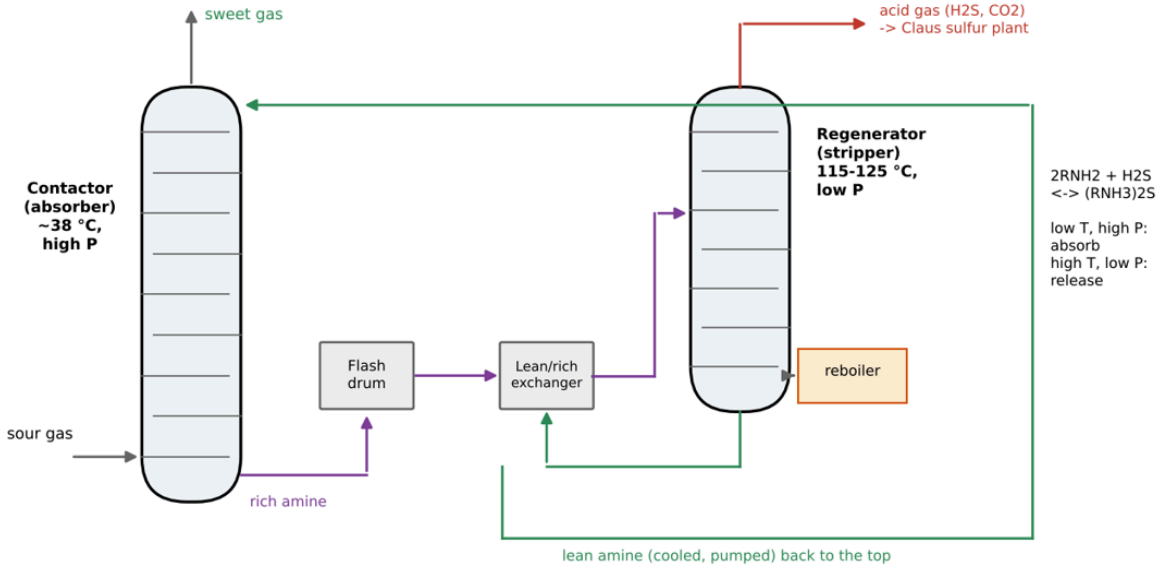


Glycol dehydration

Campbell water content of a sour gas: $W = y_{HC}W_{HC} + y_{CO2}W_{CO2} + y_{H2S}W_{H2S}$, each W read from its chart at the gas p and T (lb water per MMscf).

Sweetening (B5)

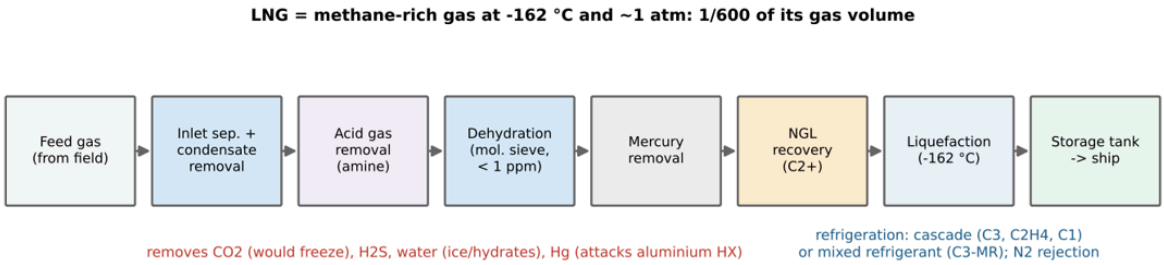
> H_2S (toxic, corrosive) and CO_2 (no heat value, freezes, corrosive) are removed by a liquid that reacts with them when cold and at high pressure and releases them when hot and at low pressure (amine), or by a solid that reacts with them (iron sponge for small H_2S).



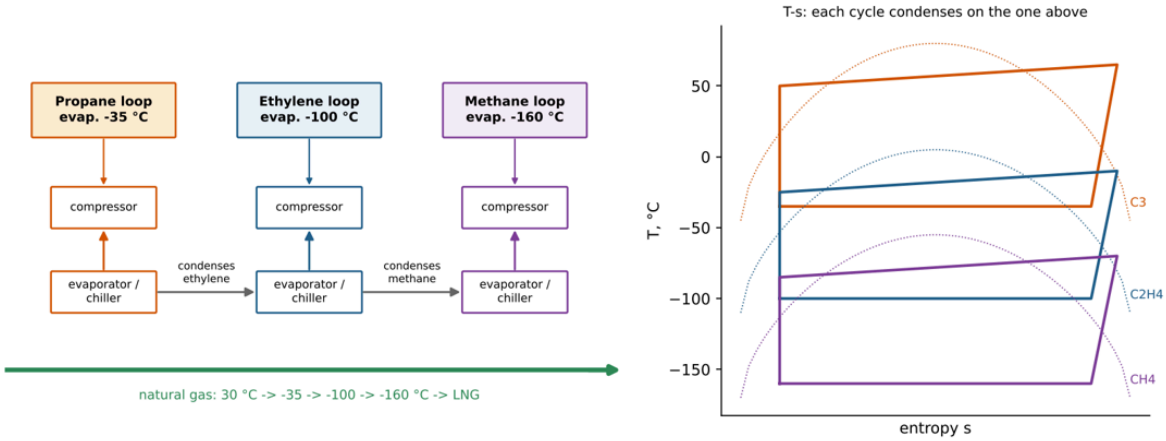
Amine sweetening

LNG (liquefied natural gas) and LPG (B6)

LNG: methane cooled to −162 °C, 600 times smaller, shipped in insulated double-hull tankers. Before cooling, everything that could freeze (CO_2, water, heavy hydrocarbons) and mercury are removed. LPG: propane + butane, liquid under small pressure, sold in cylinders; heavier than air, so leaks collect at floor level.



LNG plant blocks



Cascade refrigeration

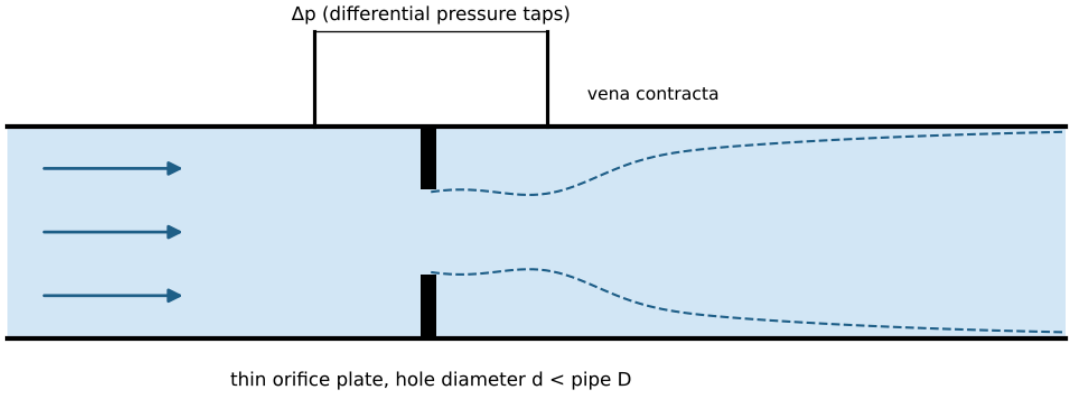
Flow through rock and gas meters (B7)

Darcy's law: flow through rock is proportional to permeability, area and pressure drop, and inversely proportional to viscosity and length. Around a well the flow is radial and pressure rises with ln r: most of the pressure is lost near the well.

$q = 0.001127\ kA(p_1 - p_2)/(\mu L)$; $p = p_{wf} + [q B\mu/(0.00708kh)]\ \ln(r/r_w)$

Symbol	Meaning	Unit
q	flow rate	bbl/day
k	permeability	md
A	cross-section area	ft^2
μ	viscosity	cp
L	length	ft
h	reservoir thickness	ft
B	oil formation volume factor	bbl/STB
r_w	well radius	ft

Example: k = 50 md, A = 1000 ft^2, Δp = 100 psi, μ = 1 cp, L = 500 ft: q = 0.001127 × 50 × 1000 × 100/(1 × 500) = 11.3 bbl/day.



Orifice meter

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
ADU	atmospheric distillation unit
AGA	American Gas Association
API	American Petroleum Institute
ASTM	American Society for Testing and Materials
bbl	barrel (42 US gallons, 5.615 ft^3)
BLEVE	boiling liquid expanding vapour explosion
BPCD	barrels per calendar day
BPSD	barrels per stream day
CBM	coal-bed methane
CCR	continuous catalyst regeneration
CDU	crude distillation unit
CNG	compressed natural gas
cp	centipoise
cSt	centistokes
DEA	diethanolamine
EFV	equilibrium flash vaporization
ETBE	ethyl tertiary-butyl ether
FCC	fluid catalytic cracking
FSRU	floating storage and regasification unit
GLR	gas-liquid ratio
GOR	gas-oil ratio
HDN	hydrodenitrogenation
HDS	hydrodesulfurization
HHV	higher heating value
KHI	kinetic hydrate inhibitor
LNG	liquefied natural gas
LPG	liquefied petroleum gas
LSR	light straight-run (naphtha)
LTS	low-temperature separator
md	millidarcy
MDEA	methyldiethanolamine
MEA	monoethanolamine
MEK	methyl ethyl ketone
MMscfd	million standard cubic feet per day
MON	motor octane number
MTBE	methyl tertiary-butyl ether

NGC	natural gas condensate
NGL	natural gas liquids
NMP	N-methyl pyrrolidone
PON	posted (pump) octane number
PSA	pressure swing adsorption
psia	pounds per square inch absolute
psig	pounds per square inch gauge
RON	research octane number
RVP	Reid vapour pressure
scf	standard cubic foot
SR	semi-regenerative (reformer); also straight-run
STB	stock-tank barrel
TAME	tertiary-amyl methyl ether
TAN	total acid number
TBP	true boiling point
TEG	triethylene glycol
VDU	vacuum distillation unit
VI	viscosity index